

**RIIM (Riset dan Inovasi untuk Indonesia Maju / Research and Innovation
for Advanced Indonesia) COMPETITION RESEARCH AND
INNOVATION PROPOSAL**



BRIN
BADAN RISET
DAN INOVASI NASIONAL



lembaga pengelola dana pendidikan

THEME

Strengthening Social and Community Resilience

TITLE

Reef to Welfare: Integrating Marine Conservation and Coastal Food-Water Security for
Sustainable Livelihoods in Indonesia's Coral Triangle (Wakatobi)

PRINCIPAL INVESTIGATOR: Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA

MEMBER: Dr. Yuna Seo

Deden Habibi Ali Alfathimy, S.H.Int., M.Si.

Eggan Nachson, M.Si.

PROPOSING INSTITUTION

Universitas Indonesia

2026

APPROVAL PAGE

RIIM COMPETITION RESEARCH AND INNOVATION PROPOSAL

1. Theme : Strengthening Social and Community Resilience
2. Proposal Title : Reef to Welfare: Integrating Marine Conservation and Coastal Food-Water Security for Sustainable Livelihoods in Indonesia's Coral Triangle (Wakatobi)
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- Research Partner Role : Kolaborasi riset internasional dalam penguatan metodologi penelitian lintas-negara dan analisis sistem sosial, mendukung pengembangan kerangka Resilience Water-Food-Climate (WFC)

5. Research Members

No	Name	Institution	Mobile/WhatsApp Number
1	Dr. Yuna Seo	Tokyo University of Science	-
2	Deden Habibi Ali Alfathimy, S.H.Int., M.Si.	National Research and Innovation Agency (BRIN)	dede029@brin.go.id
3	Eggan Nachson, M.Si.	General	+62 814-0198-1855

6. Outputs

No	Description	Year 1 (2026)	Year 2 (2027)
1	Publication in a highly reputable international journal (Q1)	1 scientific paper, status at minimum under review	1 scientific paper accepted (continuation of Year 1) + 1 new scientific paper under review
2	Food-water security training module (implementation-ready product)	Draft module & pilot project design	Final tested module + pilot project implementation report

7. Funding

No	Stage	Proposed Budget	Total Budget
1	Year 2026	Rp250.000.000	Rp250.000.000
2	Year 2027	Rp250.000.000	Rp250.000.000
	Total Proposed Budget		Rp500.000.000

This is to declare that the proposal submitted is original and has never received funding from any other institution/funding source, and does not contain any plagiarism.

Acknowledged by,

Jakarta, 30 July 2026

Head of Work Unit

Team Leader

(institutional signature and wet stamp/Digital Sign)

(signature/Digital Sign)

Prof. Debora Eflina Purba, S.S., M.Si., Ph.D.

Dr. Palupi Lindiasari Samputra, S.Pi., M.M.,
CertDA

1. TITLE

Reef to Welfare: Integrating Marine Conservation and Coastal Food-Water Security for Sustainable Livelihoods in Indonesia's Coral Triangle (Wakatobi)

2. ABSTRACT

Wakatobi embodies a development paradox: a national marine economic potential estimated at USD 1.3 trillion per year exists alongside a regional poverty rate of 14.81%, far above the national average (9.36%). As a global epicenter of marine conservation at the heart of the Coral Triangle, Wakatobi faces a dual pressure: restricted fishing access imposed by National Park zoning regulations on one hand, and the threat of climate shocks (El Niño, extreme tidal surges) that could trigger a food-water crisis on the other. This research is designed to address these issues through five interconnected stages: (1) evaluation of marine conservation governance and the sustainability of fishing households' livelihoods through spatial analysis; (2) measurement of household food security levels using the global Food Insecurity Experience Scale (FIES) standard; (3) development of a sustainable fisheries investment model based on the blue economy and circular economy; (4) analysis of the impact of climate shocks on food security and livelihoods; and (5) development of a food-water security training module as a community adaptation strategy, piloted in one small community. The research employs a concurrent mixed-methods approach with key informants from local government, customary leaders, NGOs, fishing groups, and 50 supply-chain actors, together with a quantitative survey of 300 fishing households. The novelty of this research lies in combining three domains that have thus far been studied separately — conservation governance, FIES-based micro-level food security, and circular investment models — into a single Resilience Water-Food-Climate (WFC) framework applicable to small island regional policy. The research outputs comprise publications in reputable international journals, a ready-to-use training module, and evidence-based policy recommendations for stakeholders in Wakatobi and other small island coastal regions in Indonesia.

Keywords: coastal food security; marine conservation; sustainable livelihoods; fisheries investment; climate shocks; Wakatobi

3. INTRODUCTION

3.1 Background

Indonesia possesses an extraordinary marine economic potential, reaching USD 1.3 trillion per year (KKP, 2022), yet the fisheries sector's contribution to national GDP is only around 2.54%–2.66% (KKP, 2024). This large potential, which is not matched by proportional economic benefits, has implications for the low welfare of fishers as the smallest unit of the marine social system. Of Indonesia's 3.2 million fishers, 96% are small-scale fishers (KKP, 2023). Wakatobi Regency, one of the coastal areas within Indonesia's Coral Triangle, recorded a poverty rate of 14.81% (BPS Kabupaten Wakatobi, 2023), far above the national poverty rate of 9.36%. With a population of 117,630, the number of poor households reaches 15,340, with a poverty line of Rp340.151 per capita per month — if the national rural poverty line (Rp525.000) were applied instead, Wakatobi's real poverty rate is estimated to be considerably higher. This condition is reinforced by national data showing that the Fisher's Terms of Trade (Nilai Tukar Nelayan/NTN) has continued to decline from its peak of 106.45 in 2022 to 101.76 in 2024 (BPS, 2025), far below the government's target of 108, indicating that fishers' purchasing power and welfare continue to be eroded by the cost of living and production costs.

Wakatobi's socio-economic condition illustrates a paradox of poverty amid marine wealth. Dubbed “a real underwater paradise in the center of the world's Coral Triangle”, Wakatobi is known as a world-class marine tourism destination, yet its natural wealth has not been able to bring prosperity to its population — as reflected in its poverty ranking, which is the third-highest in Southeast Sulawesi. The causes are complex: first, a weak cold chain — limited ice factories and cold storage, compounded by unstable electricity supply, causes fish quality to deteriorate before reaching markets outside the region, resulting in low selling prices. Second, fishers' dependence on middlemen due to limited cold storage forces them to sell their catch at low prices. Third, distant market access makes logistics costs very high because of Wakatobi's remote location from major consumption centers, thereby eroding fishers' profit margins.

The status of Wakatobi National Park further restricts the local-use zone compared with the core zone and protection zone, and regulations restricting fishing gear and catch quotas make investment in the fisheries sector economically less attractive than the tourism sector. Yet, according to regional economic resilience theory, dependence on a single sector

(tourism) actually signals regional economic vulnerability — residents' income becomes highly dependent on the geopolitical conditions of foreign tourists' countries of origin. Building the resilience of Wakatobi's food system requires the ecological resilience of coastal ecosystems (mangroves, seagrass meadows, salt marshes) in the face of climate change (extreme weather, sea level rise, tidal flooding). Unfortunately, periodic food security surveys in Wakatobi have not been optimally conducted, so definitive data on the region's food security level remain very limited. The issues above align with the agendas of SDG 14 (Life Below Water) and SDG 2 (Zero Hunger), and constitute the central urgency of this research.

3.2 Problem Formulation and Solution Hypotheses

RQ1: Marine conservation governance in Wakatobi potentially fails to accommodate traditional fishing access rights in a manner balanced with the goal of protecting the coral reef ecosystem.

H1 (Solution Hypothesis): Spatial analysis based on the Social-Ecological Systems (SES) and Common-Pool Resources (CPR) frameworks will identify points of convergence (and conflict) between conservation zoning and traditional fishers' fishing grounds, serving as the basis for recommendations for more participatory zoning arrangements.

RQ2: Protection of the marine ecosystem that is not integrated with coastal household food security risks creating a trade-off between conservation and welfare.

H2 (Solution Hypothesis): Measurement of household food security using the FIES (FAO) standard will produce a quantitative database that enables the integration of conservation policy with social protection schemes based on Rights-Based Approaches to Conservation.

RQ3: The scarcity of sustainable fisheries investment models causes the fisheries sector to lose out to the tourism sector in attracting investment, slowing the alleviation of fishers' poverty.

H3 (Solution Hypothesis): An investment model based on the Sustainable Livelihoods Framework, the Blue Economy, and the circular economy, with its feasibility tested through Value Chain Analysis (VCA) and Cost-Benefit Analysis (CBA), will produce an investment scheme that is both financially viable and ecologically sustainable.

RQ4: Climate shocks (El Niño, extreme tidal surges) have the potential to simultaneously cut off small island communities' access to food and clean water, yet their impact on the reef-to-welfare relationship has not been mapped at the micro/local level.

H4 (Solution Hypothesis): The water-food-climate nexus approach and measurement of the Coping Strategy Index (CSI) will reveal patterns of household vulnerability to climate shocks as a basis for designing well-targeted adaptive interventions.

RM5: Wakatobi's coastal communities do not yet have a validated food-water security training module as a coping strategy for facing climate shocks.

H5 (Solution Hypothesis): A food-water security training module developed through an Action Research approach and tested through a pilot project (quasi-experiment) will prove effective in enhancing community adaptive capacity compared to a control group.

3.3 State of the Art and Novelty

This research is structured around three research gaps identified in the prior literature. First, the ecological study by Foale et al. (2013) focused on the delimitation of marine conservation zoning but did not examine the impact of such restrictions on household-level micro food security, thereby creating the potential for spatial conflict between coral reef protection and traditional fishers' income due to the absence of FIES-based food security data. Second, the study by Pomeroy et al. (2015) identified fisheries governance (bureaucratic) issues in the Coral Triangle but did not examine a circular investment framework based on green investment as a crucial resource enabling local communities to escape poverty without sacrificing their economic access rights to capture fisheries. Third, the study on climate shocks in the Coral Triangle by Hicks et al. (2019) operates at the macro scale, whereas the direct impacts of climate shocks in fact occur at the micro/local scale — the simultaneous disruption of food access and clean water supply on small islands — which becomes significant given that communities have never been provided with the food-water security paradigm as a tangible survival strategy.

The novelty of this research lies in the integration of three domains that have thus far been studied separately — marine conservation governance, micro food security based on the global FIES standard, and circular fisheries investment models — into a single Resilience

Water-Food-Climate (WFC) analytical framework that is empirically validated at the household and community levels, while also producing a training module that can be replicated in other small island regions of Indonesia.

3.4 Objectives and Targets

Research Objectives:

1. Mengevaluasi efektivitas tata kelola konservasi laut Wakatobi dalam melindungi ekosistem terumbu karang sekaligus mengakomodasi hak akses perikanan tradisional;
2. Mengukur tingkat ketahanan pangan rumah tangga nelayan Wakatobi menggunakan standar global FIES dan mengintegrasikannya dengan kebijakan konservasi laut;
3. Mengembangkan dan menguji kelayakan model investasi perikanan berkelanjutan berbasis blue economy dan ekonomi sirkular bagi peningkatan kesejahteraan nelayan;
4. Menganalisis dampak guncangan iklim terhadap hubungan reef-to-welfare dan ketahanan pangan-air rumah tangga pesisir;
5. Menyusun dan menguji efektivitas modul pelatihan ketahanan pangan-air sebagai strategi coping komunitas pesisir melalui pilot project.

Research Activity Targets:

- a.
- b.
- c.
- d.
- e.

4. CONCEPTUAL FRAMEWORK AND STRATEGIC VALUE

4.1 Conceptual Framework

This research is built upon a series of interconnected theories to systematically address the five research questions. The effectiveness of marine conservation governance is analyzed through the Social-Ecological Systems (SES) Framework (Ostrom, 2009), which explains the interaction between the governance system (National Park regulations), actors (traditional

fishers), and the resource system (Wakatobi's coral reefs), complemented by Common-Pool Resources (CPR) theory to analyze how formal rules and customary law prevent the tragedy of the commons without marginalizing local communities. The integration of conservation and food security is explained through Rights-Based Approaches to Conservation (Campese, 2009) — the principle that nature conservation must not violate basic rights to food and livelihood — measured using the FAO Food Insecurity Experience Scale (FIES) standard, which categorizes households into severe, moderate, and mild food insecurity levels (Cafiero et al., 2018), and reinforced by the Capability Approach (Sen, 1981) to explain the root causes of food insecurity.

The sustainable fisheries investment model is built on the Sustainable Livelihoods Framework (Scoones, 1998; Allison & Ellis, 2001) — whereby household well-being is achieved through the utilization of natural, human, physical, financial, and social capital — combined with the Blue Economy (Pauli, 2010; UN Rio+20, 2012) and the circular economy (Pearce & Turner, 1990) as an investment foundation that maximizes the added value of marine resources through energy efficiency and zero waste. The impact of climate shocks is analyzed through social-ecological resilience theory (Adger, 2006) to measure the adaptive capacity of coastal communities, complemented by the water-food-climate nexus approach (Hoff, 2011; Biggs et al., 2015), which explains how climate shocks simultaneously cause loss of food access and a clean water crisis on small islands. These five theoretical frameworks are synthesized into a single, integrated analytical framework — the Resilience Water-Food-Climate (WFC) Framework — as illustrated in Figure 1.

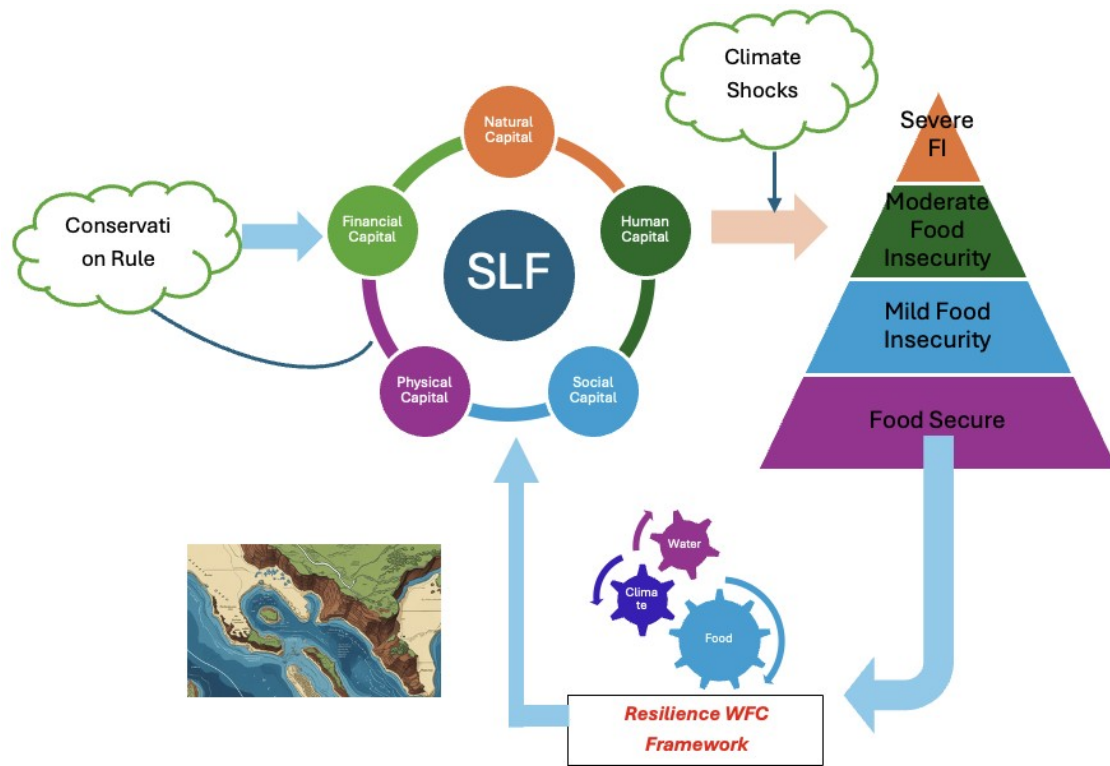


Figure 1. Resilience Water-Food-Climate (WFC) Conceptual Framework

4.2 Strategic Value

This research holds strategic value across three dimensions. Ecologically, the research findings provide recommendations for more participatory conservation governance without compromising the protection of coral reef ecosystems as a globally valuable ecological asset. Socio-economically, the research produces an accurate food security database as well as a sustainable fisheries investment model that can serve as a policy reference for alleviating fishers' poverty, in line with the achievement of SDG 2 (Zero Hunger) and SDG 14 (Life Below Water). Nationally, the resulting food-water security module has the potential to be replicated in other small island regions of Indonesia facing similar challenges, strengthening social and community resilience, which is the central theme of this research.

5. ROADMAP

The research roadmap is structured across two funding periods (2026–2027), with the final output comprising an empirically validated Resilience Water-Food-Climate (WFC) framework together with an implementation-ready training module.

Period	Activity Focus	Final Output of Period
Year 1 (2026)	Spatial analysis of conservation governance (RM1); FIES-based measurement of food and water security for 300 households (RM2); initiation of supply chain mapping and identification of investment actors (RM3)	Conservation zoning map vs. fishing grounds; household food-water security database; 1 KTI (Scientific Publication) with under review status
Year 2 (2027)	Development and feasibility testing of the sustainable fisheries investment model through VCA/CBA (RM3); analysis of climate shock impacts through CSI (RM4); development of the module and implementation of the food-water security pilot project (RM5)	Feasibility-tested investment model; climate shock impact analysis results; final training module + pilot project report; 1 KTI accepted + 1 new KTI under review

6. METHODOLOGY

6.1 Research Approach

This research employs a mixed-methods approach with a concurrent design (quantitative and qualitative components conducted in parallel) to comprehensively address the five research problem statements.

6.2 Research Location and Timeline

The research is conducted in Wakatobi Regency, Southeast Sulawesi, over a 2 (two) year funding period (2026–2027).

6.3 Methods by Research Problem Statement

Research Problem Statement	Research Method	Informants/ Respondents	Output
RM1	Qualitative: FGDs and in-depth interviews, followed by spatial analysis (GIS)	3 fisher groups; Head and technical staff of the Wakatobi National Park Office; Head of the Wakatobi Regency Marine and Fisheries Office (DKP); customary leaders (Sarano Adat); director of a local environmental NGO	Map of conservation zoning and fishers' fishing/exploration grounds

		(WWF Indonesia)	
RM2	Quantitative: survey using FIES and Water Resilience Index questionnaire instruments	300 families with a fisher as head of household or a fisher's wife as homemaker	Household food security level in Wakatobi; household water security level in Wakatobi
RM3	Quantitative: questionnaire and data analysis — Value Chain Analysis (VCA) and Cost-Benefit Analysis (CBA)	VCA/CBA respondents: 50 fisheries supply chain actors; Informants: representatives of the Regional Development Bank/fisher cooperatives, local investors/private fisheries entrepreneurs, and academics/experts in fisheries economics and circular technology	VCA and CBA calculations to determine the feasibility of sustainable fisheries investment in Wakatobi
RM4 & RM5	Mixed method: quantitative for the Coping Strategy Index (CSI) and qualitative for developing the training module, followed by Action Research (quasi-experiment)	Respondents: 300 families headed by a fisher; Informants (for the module): representatives of fisher households, the Wakatobi Regional Disaster Management Agency, representatives of the Health Office (household access to clean water), and the co-researcher team (Japan and Indonesia)	CSI measurement results; integrative food-water security training module; assessment of the success of the climate risk mitigation model through comparison of intervention and control groups

This research collaboration involves Dr. Yuna Seo (Tokyo University of Science — expertise in social systems engineering) and Deden Habibi Ali Alfathimy, S.H.Int., M.Si. (National Research and Innovation Agency) as Research Members supporting cross-country analysis and strengthening the research methodology, together with Eggan Nachson, M.Si., who is responsible for designing the AI- and OSINT-based food security system (see Appendix A — Research Team Role Format).

7. RESEARCH IMPLEMENTATION TIMELINE

The research is proposed with an implementation period of 2 (two) years of funding, namely 2026 and 2027, in accordance with the scope of the five interconnected research stages described in the Roadmap.

8. RESEARCH OUTPUTS AND PERFORMANCE INDICATORS

The targeted research outputs are set out in the Research Performance Indicators as follows:

No.	Output	Status	Target	Description
Year 2026				
1	Publication in a highly reputable international journal (Q1)	under review	1 (100%)	1 publication in a Q1 international journal with under review status, presenting the results of RM1 & RM2
2	Conservation zoning map and food-water security database	final	1 set (100%)	Spatial map + FIES and Water Resilience Index database for 300 households
Year 2027				
1	Publication in a highly reputable international journal (Q1) — continuation from Year 1	accepted	1 (100%)	1 publication in a Q1 international journal with accepted status, a continuation of the Year 1 KTI
2	Publication in a highly reputable international journal (Q1) — new	under review	1 (100%)	1 new publication in a Q1 international journal with under review status, presenting the results of RM3 & RM4
3	Food-water security training module (implementation-ready product)	final and tested	1 (100%)	Final module resulting from the pilot project (Action Research) in at least 1 coastal community in Wakatobi

In addition to the main outputs above, this research has the potential to generate additional outputs in the form of Intellectual Property (KI) over the integrated food-water security measurement method, as well as implementation of the research findings through replication of the training module in other coastal communities.

9. RESEARCH ACTIVITY SCHEDULE

No.	Activity	Implementation Period	Description
1	Research preparation, team coordination, and licensing/Research Ethics Clearance	Month 1–2 (2026)	
2	FGDs, in-depth interviews, and spatial analysis of conservation governance (RM1)	Month 2–5 (2026)	
3	FIES and Water Resilience Index survey of 300 households (RM2)	Month 4–8 (2026)	
4	Data analysis and drafting of the Year 1 scientific publication	Month 9–11 (2026)	
5	Progress report and submission of the Year 1 scientific publication	Month 12 (2026)	
6	VCA/CBA survey involving 50 supply chain actors and development of the investment model (RM3)	Month 1–5 (2027)	
7	CSI measurement and analysis of the impact of climate shocks (RM4)	Month 3–6 (2027)	
8	Development of the food-water security training module	Month 6–8 (2027)	
9	Implementation of the pilot project (Action Research/quasi-experiment) (RM5)	Month 8–10 (2027)	
10	Finalization of the module, submission of the follow-up scientific	Month 11–12 (2027)	

	publication, and final report		
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10. BUDGET

The budget breakdown for each year/funding period is prepared following the RAB (Budget Plan) Format of the 2026 RIIM Kompetisi, as shown in the RAB Proposal Details Table below, covering the components of Personnel Direct Costs, Non-Personnel Direct Costs, and Indirect Costs, with reference to the prevailing Ministry of Finance Standard Input Costs. The Rupiah values in the Volume/Frequency/Unit Price/Amount columns are to be adjusted to actual requirements prior to final submission.

RAB PROPOSAL DETAILS — YEAR 2026

Research Theme : Strengthening Social and Community Resilience

Proposal Title : Reef to Welfare: Integration of Marine Conservation and Coastal Food-Water Security for Sustainable Livelihoods in Indonesia's Coral Triangle (Wakatobi)

Principal Investigator : Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA

Home Institution : Universitas Indonesia

Research Partner : Tokyo University of Science

Total Proposed Funding Duration : 2-year funding period

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
I. DIRECT COSTS - Minimum 95% of Total Cost						
A. PERSONNEL DIRECT COSTS - Maximum 25% of Total Funding						
1	Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA (Team Leader)	5,75	1	3.600.000	Person-Month	20.700.000
2	Dr. Yuna Seo (Member)	5,75	1	2.400.000	Person-Month	13.800.000

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
3	Deden Habibi Ali Alfathimy, S.H.Int., M.Si. (Anggota)	5,75	1	2.400.000	Person-Month	13.800.000
4	Eggan Nachson, M.Si. (Member)	5,75	1	2.400.000	Person-Month	13.800.000
Total Personnel Direct Costs						62.100.000
B. NON-PERSONNEL DIRECT COSTS - Minimum 70% of Total Funding						
B.1 Marine Conservation Zoning Map vs. Traditional Fishing Grounds & Food-Water Security Database for 300 Households (FIES & Water Resilience Index)						
B.1.1 Domestic Official Travel						
1	Round-trip airfare Jakarta-Kendari-Wakatobi (research team, 3 visit phases)	5	1	2.750.000	person-trip	13.750.000
2	Speedboat rental/inter-island transportation (FGDs & field survey)	4	1	1.500.000	day	6.000.000

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
3	Team accommodation during fieldwork	9	1	450.000	Person-Day	4.050.000
4	Team daily allowance during fieldwork	9	1	300.000	Person-Day	2.700.000
Subtotal B.1.1						26.500.000
B.1.2 Field Personnel Services & Honoraria						
1	Honorarium for local enumerators/surveyors (300 FIES & WRI respondents, including enumerator training)	6	1	1.200.000	person	7.200.000
2	Honorarium for local guides & local-language interpreters	2	1	750.000	person	1.500.000
3	Survey data entry & tabulation services (300 questionnaires)	1	1	3.500.000	package	3.500.000

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
4	Spatial analysis (GIS) services for conservation zoning versus traditional fishing grounds	1	1	5.900.000	package	5.900.000
5	Statistical analysis services for FIES & Water Resilience Index data	1	1	5.300.000	package	5.300.000
Subtotal B.1.2						23.400.000
B.1.3 Consumables, FGDs, Resource Persons, Permits & Field Contingency						
1	Questionnaire printing, office/stationery supplies, document reproduction	1	1	1.800.000	package	1.800.000
2	Handheld GPS & drone rental for spatial mapping	1	1	3.800.000	package	3.800.000
3	Field data/internet package (islands with limited signal)	3	1	500.000	month	1.500.000

[illegible]

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
B.2 Publication in a Reputable Q1 International Journal (minimum status: under review)						
B.2.1 Scientific Publication Services						
1	Jasa proofreading & language editing naskah jurnal internasional Q1 [DIHAPUS - larangan Pedoman D.2.e]	0	1	7.000.000	package	0

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
2	Jasa layout/for matting & submission fee sesuai template jurnal [DIHAPUS - larangan Pedoman D.2.e]	0	1	3.000.000	package	0
Total for Sub-item B.2.1						0
Total for Sub-indicator B.2						0
B.3 AI/OSINT-Based Food-Water Security Monitoring Dashboard (third-party technical support)						
B.3.1 Third-Party Expert Services - Data Scientist & Data Analyst						
1	Data scientist services - development of the FIES and Water Resilience Index data processing pipeline	2	1	8.000.000	month	16.000.000

No.	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
2	Data analyst services - statistical analysis and visualization of survey data from 300 households	1.5	1	7.000.000	month	10.500.000
3	Development services for a machine learning-based food security (FIES) scoring model	1	1	15.000.000	package	15.000.000
4	Spatial data (GIS) integration services for the monitoring dashboard	1	1	12.000.000	package	12.000.000
5	Quality assurance and data validation services	1	1	9.100.000	package	9.100.000
Total for Sub-item B.3.1						62.600.000
B.3.2 Third-Party Expert Services - AI Developer						
1	AI developer services - architecture and backend of the OSINT monitoring system	1.5	1	10.000.000	month	15.000.000

RAB PROPOSAL DETAILS — YEAR 2027

Research Theme : Strengthening Social and Community Resilience

Proposal Title : Reef to Welfare: Integrating Marine Conservation and Coastal Food-Water Security for Sustainable Livelihoods in the Indonesian Coral Triangle (Wakatobi)

Principal Investigator : Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA

Home Institution : Universitas Indonesia

Research Partner : Tokyo University of Science

Total Proposed Funding Period : 2-year funding period

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
I. DIRECT COSTS - Minimum 95% of Total Cost						
A. DIRECT PERSONNEL COSTS - Maximum 25% of Total Funding						
1	Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA (Principal Investigator)	5,75	1	3.600.000	person-month	20.700.000
2	Dr. Yuna Seo (Member)	5,75	1	2.400.000	person-month	13.800.000

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
3	Deden Habibi Ali Alfathimy, S.H.Int., M.Si. (Anggota)	5,75	1	2.400.000	person-month	13.800.000
4	Eggan Nachson, M.Si. (Member)	5,75	1	2.400.000	person-month	13.800.000
Total Direct Personnel Costs						62.100.000
B. DIRECT NON-PERSONNEL COSTS - Minimum 70% of Total Funding						
B.1 Sustainable Fisheries Investment Model Tested for Feasibility with 50 Supply Chain Actors (VCA & CBA)						
B.1.1 Domestic Official Travel						
1	Round-trip airfare Jakarta-Kendari-Wakatobi (research team, 2 visit phases)	3	1	2.750.000	person-trip	8.250.000
2	Local/inter-island transportation (visits to 50 supply chain actors)	4	1	1.200.000	day	4.800.000

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
1	Office supplies, printing of survey instruments, documentation	1	1	2.000.000	package	2.000.000
2	FGD to validate the sustainable fisheries investment model (2 sessions)	1	1	3.000.000	FGD	3.000.000
3	Honorarium for blue economy/fisheries investment expert resource persons	1	1	2.000.000	person	2.000.000
Total for Sub-Item B.1.3						7.000.000
Total for Sub-Indicator B.1						42.450.000
B.2 Finalized and Tested Food-Water Security Training Module, CSI Measurement & Pilot Project (RM4 & RM5) + Q1 Publication						
B.2.1 Domestic Official Travel						
1	Airfare & local transportation (repeat CSI survey & pilot project implementation)	4	1	2.750.000	person-trip	11.000.000

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
2	Accommodation & team per diem during the pilot project (+/- 8 days)	5	1	750.000	person-day	3.750.000
Total for Sub-Item B.2.1						14.750.000
B.2.2 Services & Honoraria						
1	Honorarium for pilot project training facilitators	1	1	2.000.000	person	2.000.000
2	Honorarium for repeat CSI survey enumerators (300 households)	4	1	1.000.000	person	4.000.000
3	Development services for the food-water security training module (materials, curriculum, teaching media)	1	1	11.300.000	package	11.300.000
4	Climate shock impact analysis & Coping Strategy Index (CSI) services	1	1	4.700.000	package	4.700.000

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
1	Article Processing Charge (APC) publikasi jurnal internasional Q1 (status accepted, lanjutan Tahun 1) [DIHAPUS - larangan Pedoman D.2.e]	0	1	15.000.000	package	0

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
2	Jasa proofreading & language editing naskah jurnal Q1 baru (status under review) [DIHAPUS - larangan Pedoman D.2.e]	0	1	6.000.000	package	0
Total for Sub-Item B.2.4						0
Total for Sub-Indicator B.2						42.850.000
B.3 Enhancement of the AI/OSINT-Based Food-Water Security Monitoring Dashboard (third-party technical support)						
B.3.1 Third-Party Expert Services - Data Scientist & Data Analyst						
1	Data scientist services - advanced VCA/CBA data analysis & dashboard integration	1.5	1	8.000.000	month	12.000.000

No	Research Cost Component / Activity / Justification of Need	Volume	Frequency	Unit Price (Rp)	Unit	Amount (Rp)
2	Data analyst services - Coping Strategy Index (CSI) analysis & climate shock impact	1.5	1	7.000.000	month	10.500.000
3	Development services for a predictive model of climate impact on food security	1	1	14.000.000	package	14.000.000
4	Advanced data integration & synchronization services to the dashboard	1	1	10.000.000	package	10.000.000
5	Advanced quality assurance & data validation services	1	1	6.100.000	package	6.100.000
Total for Sub-Item B.3.1						52.600.000
B.3.2 Third-Party Expert Services - AI Developer						
1	AI developer services - refinement of the AI/NLP model for OSINT monitoring	1	1	14.000.000	package	14.000.000

11. REFERENCES

Prepared in accordance with American Psychological Association (APA) referencing style.

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APPENDIX A. RESEARCH TEAM ROLE FORMAT

RESEARCH TEAM COMPETENCE

Describes the team structure, competencies, roles, and experience relevant to the implementation of the activity.

No.	Name	Education	Expertise	Role in Research Activities
1	Dr. Palupi Lindiasari Samputra, S.Pi., M.M., CertDA	Doctoral (S3)	Microeconomic Resilience / Marine and Fisheries Economics	Principal Investigator — overall research coordination, design of the food security-economic analytical framework, and analysis of sustainable fisheries investment models (RM3)
2	Dr. Yuna Seo	Doctoral (S3)	Social Systems Engineering	Member/Co-Researcher — strengthening of cross-country methodology and social systems analysis within the Resilience Water-Food-Climate framework (supporting RM1 & RM4)
3	Deden Habibi Ali Alfathimy, S.H.Int., M.Si.	S2	Kebijakan Publik (Kebijakan Kedingrintaraan)	Member/Co-Researcher — cross-institutional research collaboration with the National Research and Innovation Agency, strengthening of policy analysis (supporting RM1, RM2, & RM3)
4	Eggan Nachson,	Master's (S2)	Artificial	Research Member

	M.Si.		Intelligence and Data-Driven Business	(Research Assistant) — design of an AI- and OSINT (Open Source Intelligence)-based community food security system; design of dashboards, policy briefs, and monthly reports on regional food security conditions (supporting RM2, RM4, & RM5)
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